How far the answer moves: an RGB-distance readout beside expected exact match

> We re-scored ex-2.2.11's 54 checkpoints with a distance on the color grid beside expected exact match, to see where the answers go when the anchored direction is removed. On `handover` the removed answers move 44% to 77% of the way from a perfect answer to a random guess, and the non-red lines move less than the control's own lines do under the same operator.

Observations

- The removal lines, by distance: under `projection`, the greedy guess on `handover` moves from the floor (0 to 0.5 steps, a perfect answer) to 1.8 steps on `mix` and 3.4 on `value-hsv`, between 44% and 77% of the way to chance. The control stays at the floor. The mean, median, and expected distances agree with the greedy one within 0.3 steps.
- Kept and lost: the operator loses 77% to 99% of the removal lines. The kept lines stay at the floor; the lost lines sit 1.8 (`mix`) to 3.7 (`value-hsv`) steps from the answer, and reach chance only on `value-hsv`.
- The lost answers on the cube: on `mix`, nine in ten lost answers are one or two steps off. On `hue-hsv` the guesses keep their saturation and move in hue; on `darken` they land darker than the answer.
- A counterfactual answer: the lost answers are not the op applied to a de-reddened operand. The nearest such answer is a little closer to the guess than the true one on most ops, but the guess lands in its box on under a quarter of the lines on all ops but `lighten`.
- The non-red lines: on `handover` the exact match deficit stays under 0.01 and the mean of the distribution moves under 0.03 steps on every op. The control's own lines move as much or more under the same operator.
- Between seeds: the seed spread of the distances is 6% to 20% of the mean, against 33% to 127% for the kept share of exact match.

Scope

This is a re-score of stored checkpoints, planned in the backlog item. There is no training, no gates, and no verdicts.

Every removal and selectivity number in D2.2 is *expected exact match*: the probability mass the model puts on the colors that a line's answer can be. It gives no partial credit, so a guess one grid step from the answer scores the same as a guess of black.

So when `handover` keeps 24% of exact match on the `hue-hsv` removal lines, that could mean a quarter of the lines answered perfectly and the rest sent anywhere, or every line nudged one step over. Likewise, a non-red deficit of 0.01 could be one line in a hundred lost outright, or the mass of every line leaning a little.

We add a distance on the color grid, in grid steps. One step is the spacing between the six levels of each channel, and the far corner of the cube is $5\sqrt{3} \approx 8.7$ steps from black. The target of a line is its *raw answer* (the unrounded value of the op), which is also the expected value of the answer distribution under stochastic rounding.[1]

Four quantities are measured from the raw answer. Three are central points of the model's distribution over the 216 grid colors: the *greedy* guess (the most likely color), the *mean*, and the channel-wise *median*. The fourth is the *expected* distance of a color drawn from the distribution.

The greedy and expected distances are the ones to compare with exact match. The mean is the one that moves when mass leans toward a neighbor while the guess stays the same.

Every distance comes with two references. The *floor* is the distance from the raw answer to the best possible grid answer (the mode of the answer distribution): zero when the answer lands on the grid, and up to half a step per channel otherwise. A distance at the floor is a perfect answer.

Chance is the mean distance of a color drawn uniformly from the grid: about 3 steps for a target near the middle of the cube, and more for one near a corner. A distance at chance means the answer no longer depends on the line.

The distances and exact match are measured under the same three passes as ex-2.2.11: (a) the clean forward pass, (b) `projection`, with e_1 removed at every slice and position, and (c) `operands`, with e_1 removed at the operand positions only. Each is read on the line groups of ex-2.2.11: the removal lines (a red operand whose hue the op carries into the answer), the non-red lines, and the rest. As in the ex-2.2.11 report, the shaped operator is left out.

The runs

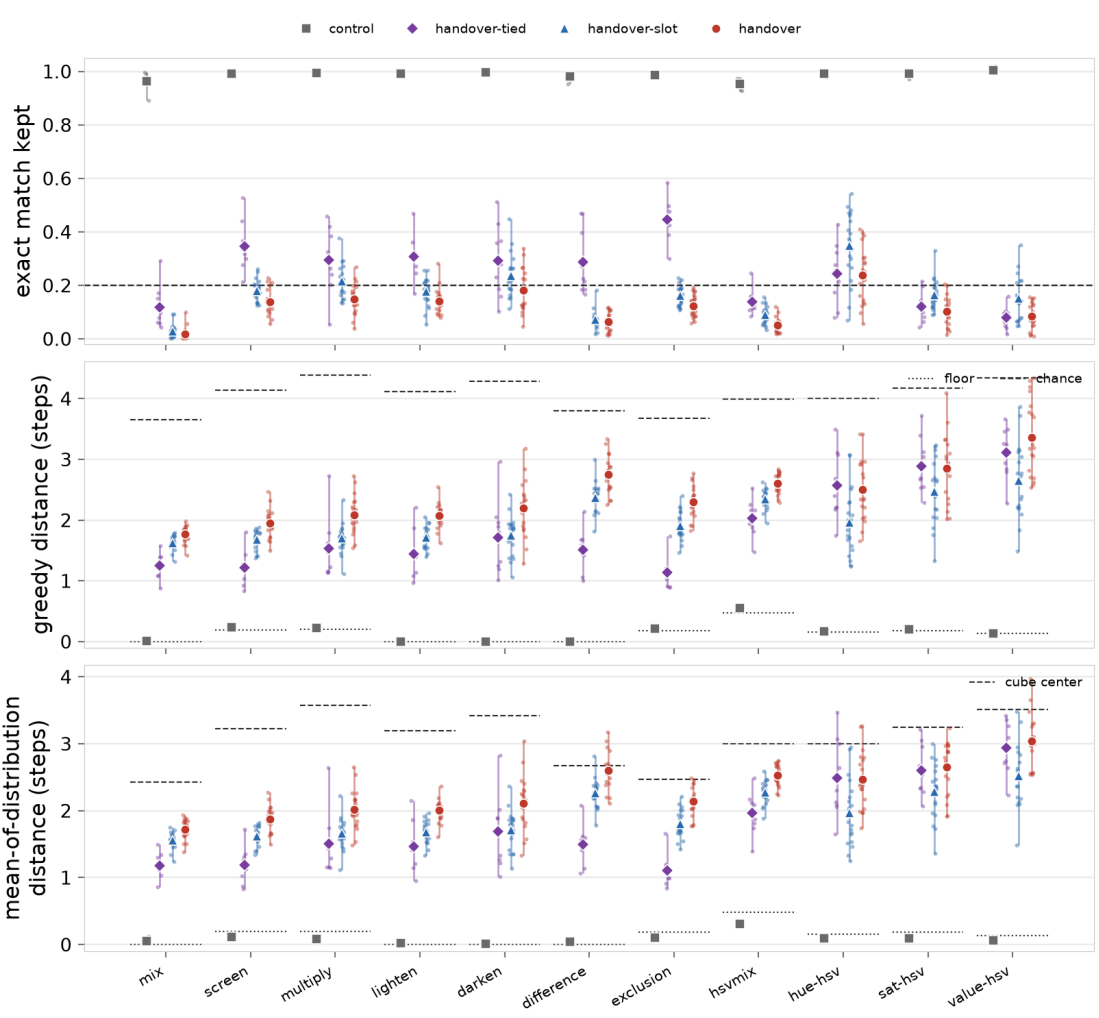
The runs are the 54 final checkpoints of ex-2.2.11: the un-anchored control (5 seeds), `handover` (20), and two arms that each undo one of the changes in `handover`, `handover-slot` (20) and `handover-tied` (9). The probe lines are the ones ex-2.2.11 scored: 5,832 per op, and twice that for the three order-sensitive ops.

The two scorers agree: the re-score reproduces ex-2.2.11's kept share of expected exact match on every op (24% on `hue-hsv`, 12% to 18% on the next four).

The removal lines, by distance

On the removal lines under `projection`, the greedy guess on `handover` moves about halfway from the floor to chance on six ops (44% to 51% of the way), and further on `difference`, `exclusion`, and the three HSV ops (61% to 77%). The figure puts the distance beside the kept share, per op.

The clean pass sits at the floor on every op and condition (the table below the figure gives it), so any distance above the floor under `projection` is caused by the operator.



The removal lines under `projection`, per op and condition: exact match kept, and how far the answer moves. *Top: the share of the clean expected exact match kept, as ex-2.2.11 reads it, dashed at its 20% gate. Middle: the distance in grid steps of the greedy guess from the raw answer; the dotted rule at each op is the floor (the best grid answer's distance) and the dashed rule is chance (a uniform draw from the grid). Bottom: the distance of the mean of the model's distribution from the raw answer; the dashed rule is the center of the cube, where a uniform distribution's mean sits. Each small dot is one seed, the larger mark the seed mean, and the thin bar the seed range.*

A sampled answer lands about as far as the greedy one. Sampling at temperature 1 draws from the model's distribution, so the expected distance is what it scores: 0.03 to 0.18 steps above the greedy distance on every op (the table gives it). The greedy guess is the limit as the temperature goes to zero. The mean of the distribution describes no single sample; it is the summary that moves when mass leans toward a neighbor while the guess stays put, and it sits a little below the greedy distance.

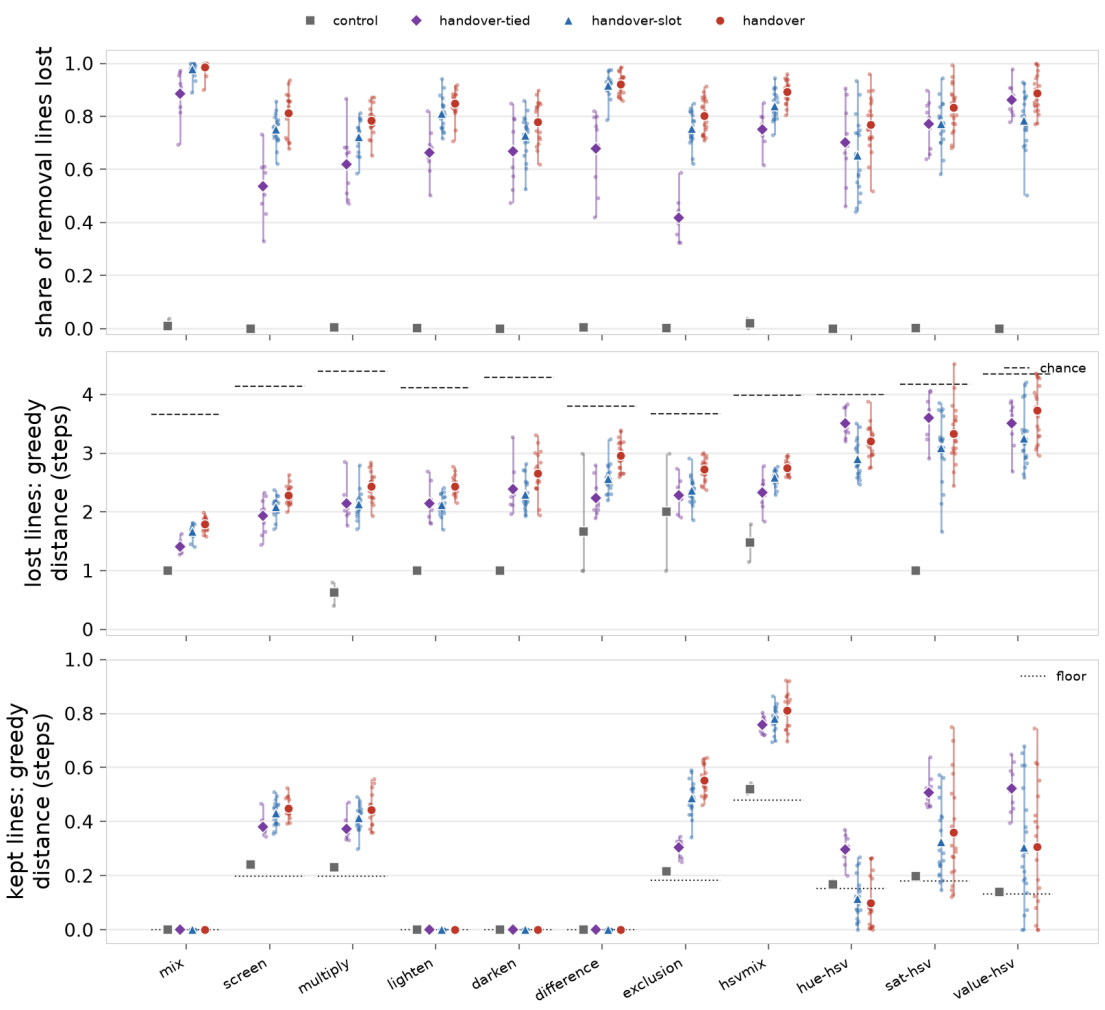
The table gives the same measurements on `handover`, adds the clean and `operands` passes, and adds the hit rate (the share of lines whose greedy guess is one of the line's possible answers).

op	removal lines	exact match kept	hit rate, clean	greedy distance, `projection`	greedy distance, clean	greedy distance, `operands`	expected distance, `projection`	floor	chance
`mix`	365	0.02 (0.0–0.1)	1.00 (1.0–1.0)	0.01 (0.0–0.1)	0.00 (0.0–0.0)	1.77 (1.4–2.0)	1.76 (1.4–2.0)	1.87 (1.5–2.1)	0.00	3.66
`screen`	387	0.14 (0.1–0.2)	1.00 (1.0–1.0)	0.19 (0.1–0.3)	0.24 (0.2–0.2)	1.95 (1.5–2.5)	1.94 (1.5–2.4)	2.06 (1.7–2.5)	0.20	4.14
`multiply`	384	0.15 (0.0–0.3)	0.91 (0.9–0.9)	0.15 (0.1–0.3)	0.22 (0.2–0.2)	2.09 (1.5–2.7)	2.07 (1.5–2.8)	2.17 (1.6–2.8)	0.20	4.39
`lighten`	386	0.14 (0.1–0.3)	1.00 (1.0–1.0)	0.15 (0.1–0.3)	0.00 (0.0–0.0)	2.07 (1.6–2.6)	2.07 (1.6–2.5)	2.14 (1.7–2.6)	0.00	4.12
`darken`	382	0.18 (0.0–0.3)	0.95 (0.9–0.9)	0.17 (0.0–0.3)	0.00 (0.0–0.0)	2.20 (1.3–3.2)	2.18 (1.3–3.2)	2.23 (1.4–3.1)	0.00	4.28
`difference`	403	0.06 (0.0–0.1)	0.99 (1.0–1.0)	0.07 (0.0–0.1)	0.01 (0.0–0.0)	2.75 (2.3–3.3)	2.73 (2.2–3.3)	2.84 (2.3–3.4)	0.00	3.80
`exclusion`	390	0.12 (0.1–0.2)	1.00 (1.0–1.0)	0.20 (0.1–0.3)	0.21 (0.2–0.2)	2.30 (1.8–2.8)	2.25 (1.8–2.8)	2.35 (1.9–2.7)	0.18	3.67
`hsvmix`	401	0.05 (0.0–0.1)	0.98 (1.0–1.0)	0.09 (0.0–0.2)	0.53 (0.5–0.6)	2.61 (2.3–2.8)	2.59 (2.2–2.8)	2.70 (2.4–3.0)	0.48	3.99
`hue-hsv`	333	0.24 (0.1–0.4)	1.00 (1.0–1.0)	0.23 (0.0–0.5)	0.17 (0.2–0.2)	2.51 (1.7–3.4)	2.38 (1.7–3.4)	2.56 (1.8–3.4)	0.15	4.00
`sat-hsv`	377	0.10 (0.0–0.2)	1.00 (1.0–1.0)	0.17 (0.0–0.3)	0.20 (0.2–0.2)	2.85 (2.0–4.1)	2.71 (2.0–3.4)	3.03 (2.2–3.7)	0.18	4.17
`value-hsv`	389	0.08 (0.0–0.2)	1.00 (1.0–1.0)	0.11 (0.0–0.2)	0.15 (0.1–0.2)	3.36 (2.5–4.3)	3.15 (2.1–3.9)	3.38 (2.8–4.2)	0.13	4.34

The removal lines on `handover`, per op: seed mean (seed range). *Distances in grid steps from the raw answer; the expected distance is what a sample at temperature 1 scores. The floor is the best grid answer's distance and chance a uniform draw's, both averaged over the op's removal lines.*

Kept and lost

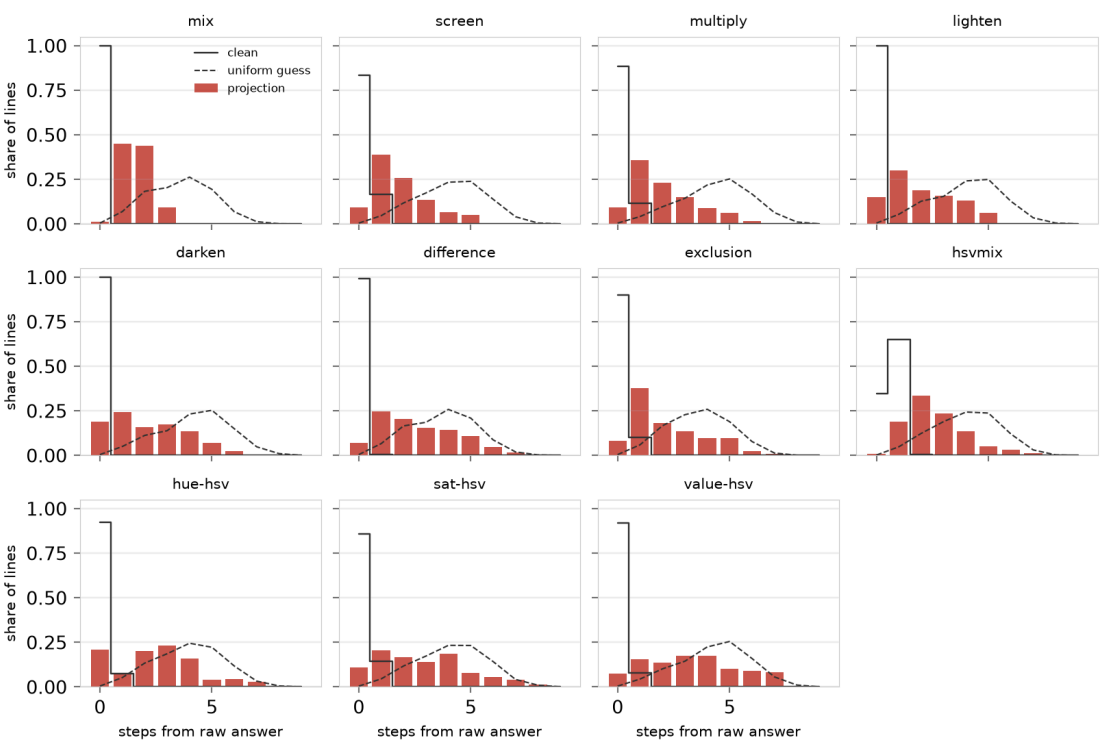
A distance averaged over a group mixes lines the operator leaves answered with lines it does not, so we split the lines by the greedy guess. Under a given pass, a line is: (a) *kept* when the guess is one of the possible answers of the line, (b) *lost* when the clean guess was a possible answer and the guess under the pass is not, and (c) *never* when the clean guess already missed. The distance of the lost lines tells us where a removed answer goes.



The removal lines under `projection`, split by whether the greedy guess survives. *Top: the share of each op's removal lines that are lost (answered on the clean pass, not under the operator). Middle: the greedy distance of the lost lines from the raw answer, in grid steps, with chance dashed. Bottom: the greedy distance of the kept lines, with the floor dotted. Each small dot is one seed, the larger mark the seed mean, and the thin bar the seed range; a condition with no lost line on a seed draws nothing for it.*

Per line, as a distribution: in the histograms below, the bin at zero holds the kept lines and the other bins show where the lost answers go.

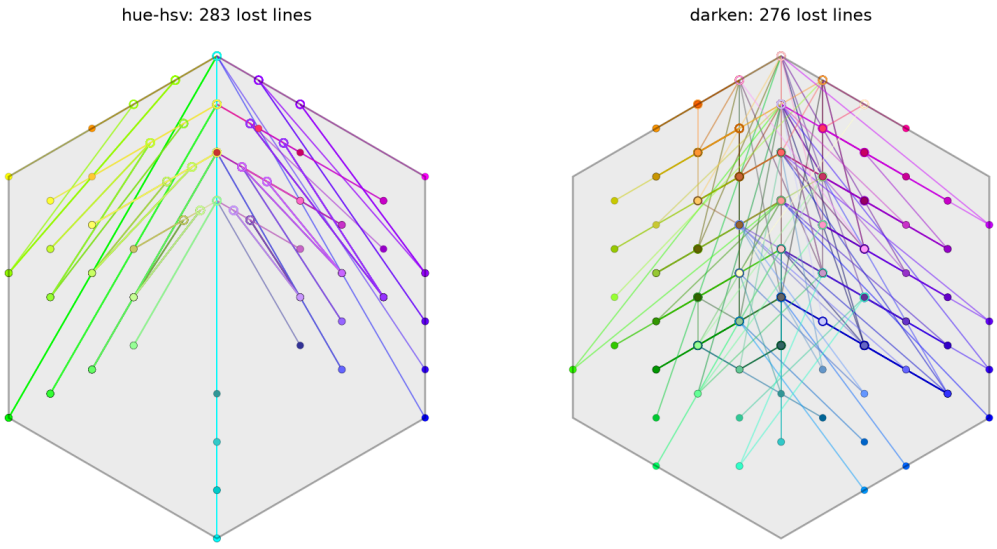
On `mix`, nine in ten of the moved answers sit one or two steps off, which is what treating the red channel as absent would give (on `mix` that shift is about 2.5 steps). `darken` decays from one step, `hue-hsv` spreads over two to four, and `value-hsv` is close to the shape a uniform guess gives.



Where the removal-line answers land under `projection` on `handover`, per op. *Bars: the share of the op's removal lines, pooled over twenty seeds, whose greedy guess sits each number of grid steps from the raw answer (rounded), under `projection`; in outline, the same under the clean pass. The dashed line is the distribution a uniform guess from the grid would give on the same lines.*

The lost answers on the cube

The figure draws the lost removal lines of the two ops the backlog item names: `hue-hsv`, whose removal lines keep the most exact match, and `darken`, the near miss of ex-2.2.11.



Where the lost removal lines land, on `handover-s0` under `projection`.

One panel per op, on the color wheel view of the cube. Each mark is the greedy guess of one lost line, in the guessed color; the ring is the line's raw answer and the stub joins the two. A random sample of the lost lines is drawn where there are more than fit.

A counterfactual answer

This is a post hoc check on where the lost answers go. If the operator took the redness out of the red operand and left the op intact, the guess would be the op applied to a de-reddened operand. We try two versions of that operand: (a) red channel set to zero, and (b) red channel lowered to the larger of its green and blue, which keeps the other two channels and drops the saturation.

op	to the answer	to the zero-red answer	to the grey-red answer	in the grey-red box
`mix`	1.77	1.98	1.67	5%
`screen`	1.95	2.32	1.94	17%
`multiply`	2.09	2.32	2.02	22%
`lighten`	2.07	1.68	1.52	36%
`darken`	2.20	3.03	2.55	10%
`difference`	2.75	2.90	2.69	5%
`exclusion`	2.30	2.45	2.14	20%
`hsvmix`	2.61	2.41	2.22	3%
`hue-hsv`	2.51	2.79	2.03	21%
`sat-hsv`	2.85	4.56	4.14	0%
`value-hsv`	3.36	3.77	2.89	8%

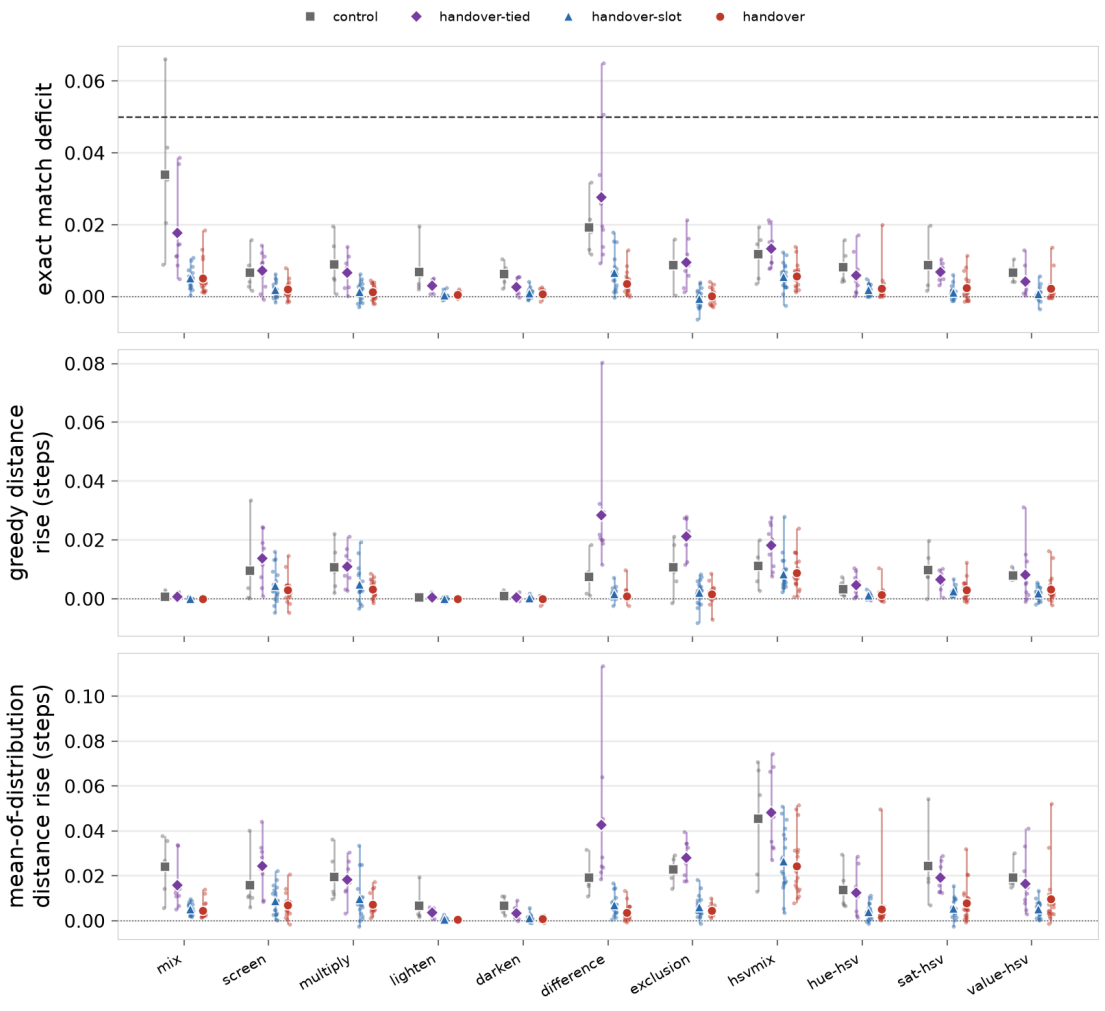
Distance of the greedy guess on the removal lines, `handover` under `projection`, seed mean. *The answer is the line's raw answer; the zero-red and grey-red answers are the op applied with the red operand's red channel set to zero, or to the larger of its green and blue. The last column is the share of lines whose guess is one of the grey-red answer's possible roundings.*

Neither substitute is where the guesses go. The zero-red answer is farther from the guess than the true answer on every op but `lighten` and `hsvmix`. The grey-red answer is a little nearer on most ops (2.0 steps against 2.5 on `hue-hsv`), but the guess lands in its box on under a quarter of the lines on every op but `lighten` (36%).

So the lost answers sit between the answer and its de-reddened version, and no single substitute operand describes them.

The non-red lines

The non-red lines are the other side of selectivity: the operator should leave them alone. Ex-2.2.11 gates the exact match deficit on `mix` at 0.05, and finds it near zero on every op. The distance shows whether the mass moves at all on those lines.



The non-red lines (without a red answer) under `projection`, per op and ***condition: what the operator changes.*** *Top: the deficit in expected exact match (clean minus `projection`), dashed at ex-2.2.11's 0.05 gate, which that report reads on `mix` alone. Middle: the rise in the greedy distance from the raw answer, in grid steps. Bottom: the rise in the distance of the mean of the model's distribution. Each small dot is one seed, the larger mark the seed mean, and the thin bar the seed range.*

Between seeds

The distances vary far less between seeds than exact match does. A measurement used for a gate needs a seed spread small relative to the effect it measures; the table gives each removal measurement's spread across the twenty `handover` seeds as a share of its mean (the coefficient of variation, CV).

op	exact match kept: mean	seed CV	greedy distance: mean	seed CV	mean-of-distribution distance: mean	seed CV
`mix`	0.02	127%	1.77	8%	1.73	9%
`screen`	0.14	35%	1.95	12%	1.88	10%
`multiply`	0.15	38%	2.09	15%	2.01	15%
`lighten`	0.14	33%	2.07	10%	2.00	8%
`darken`	0.18	44%	2.20	20%	2.11	19%
`difference`	0.06	56%	2.75	11%	2.60	11%
`exclusion`	0.12	35%	2.30	11%	2.14	10%
`hsvmix`	0.05	53%	2.61	6%	2.53	7%
`hue-hsv`	0.24	44%	2.51	20%	2.46	17%
`sat-hsv`	0.10	53%	2.85	19%	2.65	15%
`value-hsv`	0.08	58%	3.36	18%	3.04	13%

Seed spread of the removal measurements on `handover` under `projection`, per op. *CV is the standard deviation over the twenty seeds divided by the seed mean.*

What we make of it

Under `projection`, nearly every removal line moves, and moves part of the way: the greedy guess ends about halfway from a perfect answer to a random guess on six of the eleven ops, and further on the rest. On `mix`, the op the D2.2 gates are scored on, the guess ends one or two steps from the answer nine times in ten.

The answers move toward what the line would give without its red, but not all the way: the counterfactual check says the model is not computing the op on a de-reddened operand.

That is the kind of side-effect this program wants to be able to describe. The intervention changes the answer on the lines that use the concept, by an amount that scales with how much the op relies on the red channel, and leaves the other lines alone.

The distance gives the removal measurement partial credit and a scale. On `mix`, exact match on `handover` sits near zero with a seed spread larger than its mean, while the distance on the same lines has a seed spread under a tenth of its mean.

That comparison flatters the distance: the CV favors a measurement with a large mean, and the distances have means an order of magnitude above the kept shares. In units of the seed spread, the two separate the conditions about equally well. The advantage of the distance is that it stays informative where exact match is near zero.

The distance also separates ops that exact match does not: `mix` and `value-hsv` both keep under a tenth of their exact match, but their guesses sit 1.8 and 3.4 steps away.

On the non-red lines, the mean of the distribution is the sensitive measurement, since it moves while the guess stays put. On `handover` every such move is under 0.03 steps, and the control's lines move as much or more, so the operator's cost on the lines it should leave alone is about what removing any direction costs.

We adopt the distance for D2.2's removal and selectivity gates, from the next experiment that scores them, with this direction: (a) removal, as the normalized distance of the mean of the distribution on the removal lines, (distance − floor) / (chance − floor), rising under the operator, with the greedy distance beside it as the number a user of exact match would expect; and (b) selectivity, as the non-red rise of the same mean staying at or under the rise of the control under the same operator. The thresholds are the one thing a re-score cannot supply, since it has no gate to calibrate against, so that experiment sets them from its own control runs. Until one of the distances has been used in a gate, the greedy and mean distances are quoted together.

The distance does not say whether an answer is right, except at the floor, so a gate that wants "answered" should keep the hit rate or exact match beside it.

Method

The distances are in `sca.answer_distance`, and the re-score in `experiment.py` beside this script. Each of the 54 final checkpoints of ex-2.2.11 is run over the same probe lines under the clean pass, `projection`, and `operands`, with one compiled forward pass per operator. The answer distribution at the pre-answer position is restricted to the 216 color tokens.

For each line, we compute against its raw answer: the greedy guess, the mean and channel-wise median of that distribution, its expected exact match and expected distance, and whether the guess is a hit. For each group and pass, these are averaged, split by kept, lost, and never, and binned by rounded greedy distance. On `handover` and the control, the per-line guess is published for every op, and the mean beside it for `hue-hsv` and `darken`; every other per-line array stays in the run's memo record.

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1. Stochastic rounding picks the grid level above or below a value at random, weighted so that the average equals the unrounded value. ↩